ENSEMBLE SYSTEM FOR ROBO-INSTRUMENT PLAYER AND EXPERIENCED MUSICIAN USING SCORE FOLLOWING

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ABSTRACT

We develop a system that enables a beginner of musical performance to easily participate elaborated ensemble performances with an experienced musician. This system is composed of two key elements: a robo-instrument, which enables the beginner player to take a part in the ensemble by automatic mechanical controls of fingering motions for musical instruments such as saxophone, and a realtime music alignment (score following) method that takes the performance signal of the experienced musician as input and estimates the current position in the performed musical score. To realize a seamless and flexible ensemble, we study a MIDI-based score following method that can treat performance errors, repeats, and large skips, and a method for predictive transmission of MIDI events to the robo-instrument. We demonstrate this system with pieces for piano and saxophone, where the latter part is played with a robo-saxophone, and discuss new problems of music information processing required for extending the research on such realtime interactive systems.

1. INTRODUCTION

Musical ensemble performance offers a compelling context for fostering social bonds [1, 2]. However, acquiring the instrumental skills required for engaging in satisfying ensemble performance is often challenging, particularly for beginners. One promising approach to addressing this difficulty is the use of robo-instruments [3], which are semi-automated devices that enable users to play real acoustic instruments with minimal training, without the need to master complex motor skills such as fingering. Because fingering in robo-instruments is controlled by MIDI signals, it is possible to synchronize robo-instruments to enable coordinated ensemble performance. A key technical challenge in realizing such an ensemble lies in synchronizing the MIDI signal with a co-player's expressive performance, which naturally involves tempo fluctuations, performance errors, and, in rehearsal-like situations, repeats and skips to arbitrary positions in the score.

Figure 1. Robo-saxophone.

To develop an ensemble system with which a beginner performs one part (hereafter, the follower part) with a robo-instrument and an experienced musician performs another part (hereafter, the leader part) with a MIDI-equipped musical instrument, we investigate an adaptation of score following techniques [4, 5]. To compensate for the latency in the motor controls of the robo-instrument, we propose a method for predictive transmission of MIDI events by estimating the local tempo of the leader part in real time.

In the demonstration, we will present an ensemble system with a robo-saxophone and a digital piano, allowing conference attendees to experience the realtime interaction enabled by the system. We will also discuss several technical challenges associated with realtime interactive systems of this kind, including aspects of machine learning, control theory, and cognitive modeling.

2. ROBO-INSTRUMENT

Robo-instruments are semi-automatic devices that automates part of the instrumental performance through mechanical control [3]. Figure 1 shows an example of a robo-saxophone. This system is constructed by mounting 19 servo motors, a microcontroller, and a servo driver onto an alto saxophone, with all components fixed using 1.6 mm thick metal plates. The servo motors pull wires to press the keys via rotational motion. Since the keys are actuated automatically by the machine, the performer can play the instrument by simply blowing into the mouthpiece.

The system operates on a 5V power supply and is powered by a lithium-ion battery. It is controlled via MIDI, allowing the instrument robot to function as an external MIDI sound source. The key actuation time is approximately 30 ms, which is comparable to or faster than that of a human performer.

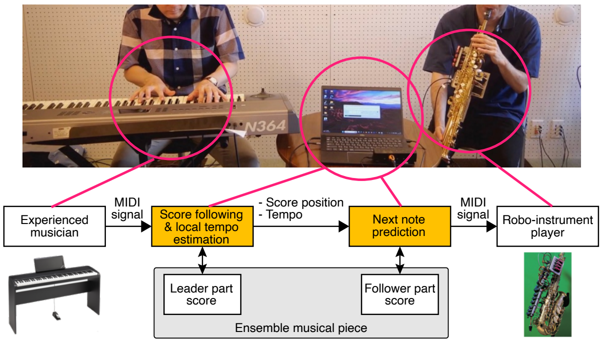

Figure 2. The proposed ensemble system.

3. SYSTEM ARCHITECTURE

Figure 2 illustrates the proposed ensemble system. In this system, an experienced musician performs a MIDI-capable instrument (e.g. a digital piano) in ensemble with a beginner who plays a robo-instrument. The ensemble follows a pre-registered musical score consisting of a leader part and a follower part. The system comprises two main components: (i) a score-following module, which estimates the leader's current position in the score in real time, and (ii) a MIDI signal generation module, which transmits MIDI events to control the robo-instrument. To compensate for the actuation delay inherent in the robo-instrument and latencies in data processing including score following, the system also estimates the leader part's local tempo in real time and transmits MIDI events for the follower part in advance, based on next note prediction.

4. SCORE FOLLOWING AND MIDI SIGNAL GENERATION

A MIDI-based score-following algorithm is employed, based on the outer-product hidden Markov model [6]. This method can handle polyphonic performance and is robust to insertion, deletion, and substitution errors in note events, as well as arbitrary repeats and skips. To estimate the local tempo from the score-following results, a moving average method is used. With each incoming note from the leader part, the system updates three internal values in real time: (1) the current position in the score, (2) the corresponding upcoming notes for the follower part, and (3) the predicted onset time for these notes.

The actual transmission of MIDI events for the follower part is carried out in advance of the predicted onset time. A tunable parameter, denoted as the lead time Δ, is introduced to control this process. During operation, as the leader part's note events are sequentially processed and the predicted onset times of the follower part are updated accordingly, additional logic is required to maintain smooth ensemble coordination. Specifically, if the difference between the current prediction and the previous one is within a predefined threshold (e.g., 1 s), MIDI transmission for the following part is temporarily suppressed until the predicted onset time of the next note. When the prediction shift exceeds this threshold, new MIDI events for the lower part are sent based on the updated timing. This mechanism enables the system to handle large repeats and skips made in the leader part, while avoiding redundant MIDI transmission under local tempo fluctuations, thereby facilitating smoother ensemble performance.

5. EXPERIMENT AND INFORMAL EVALUATION

The system was implemented and evaluated to verify its functionality and to investigate suitable values of the lead time Δ for enabling smooth ensemble performance. In the experiment, a pianist with over 30 years of performance experience played a digital piano, while a robo-saxophone was played by a participant with no knowledge of saxophone fingering and no ability to read musical scores. An arrangement of Chopin's Nocturne Op. 9, No. 2 was used.

The accompanying demo video, recorded during the experiment, shows that the robo-instrument player was able to perform seamlessly in ensemble with the pianist, even in the presence of tempo fluctuations and unplanned repeats. The experiment revealed that a suitable value for the lead time Δ was approximately 80 ms. However, the optimal value varied depending on the musical context such as the duration, articulation, and phrasing of individual notes in the follower part. For example, in the case of short ornamental notes played without tonguing, it was preferable to set the lead time just long enough to compensate for the actuation and data processing delays. In contrast, for longer notes or notes at the beginning of a phrase, a relatively longer lead time was found to be preferred, as it allowed the player to exert more nuanced control over embouchure and breathing.

6. DISCUSSIONS

We developed an ensemble system that enables a beginner player to engage in a satisfying ensemble with an experienced musician. Our experiment confirmed that the integration of a state-of-the-art MIDI-based score following with predictive MIDI signal generation based on realtime local tempo estimation allows for seamless and flexible ensemble performance.

The results also highlighted several technical challenges to further improve the usability of such realtime interactive systems. Addressing these problems may require a multidisciplinary approach involving machine learning (ML), control theory (CT), and cognitive modeling (CM).

- Optimization of lead times for individual notes based on their musical contexts (ML, CM).
- Development of methods allowing beginner players to partially control their own tempo for enabling ensemble with mutual interaction (CT, CM).
- Integration of score following with acoustic input [7, 8], enabling performances by experienced musicians using non-MIDI-equipped instruments (ML).

7. ACKNOWLEDGEMENT

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